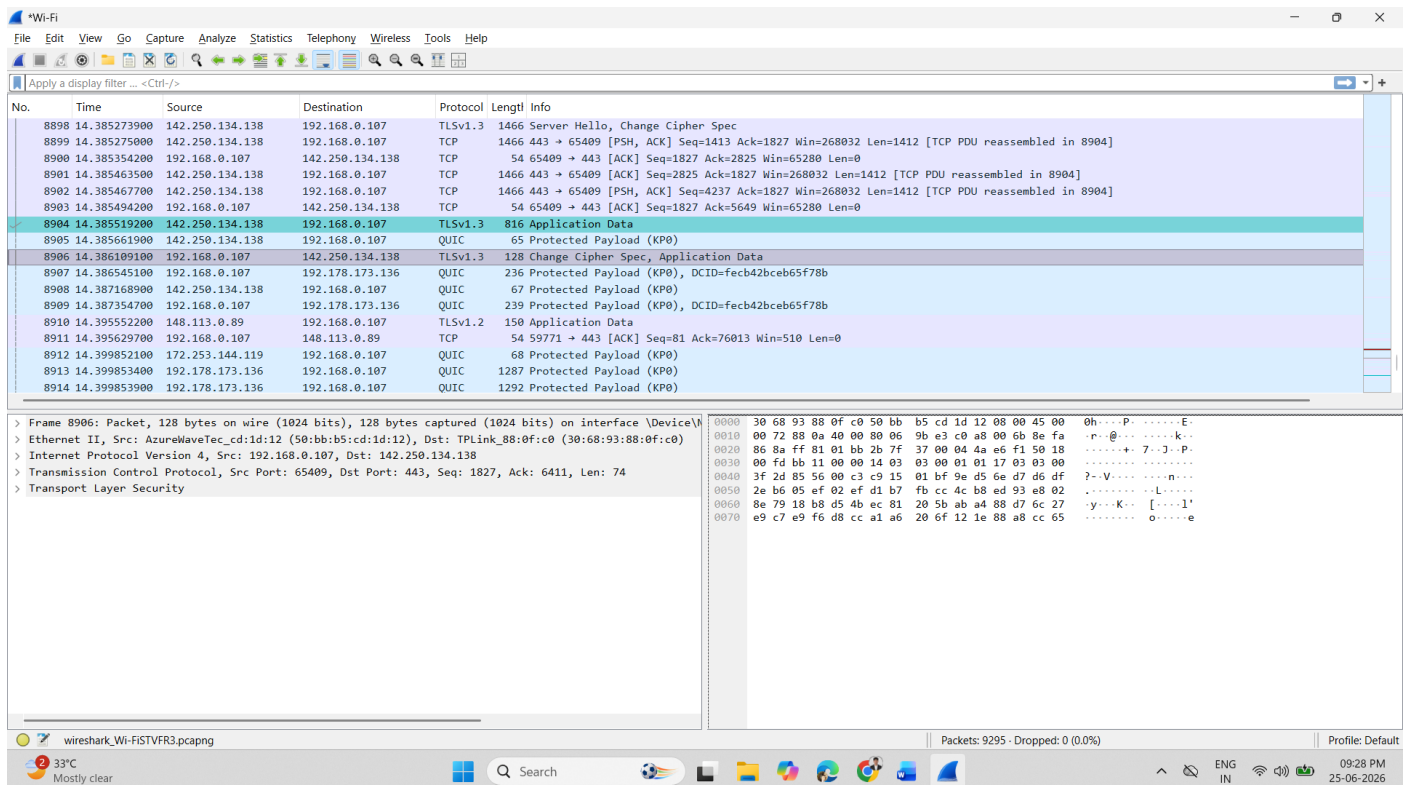


Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

ANS :

- -After installing Wireshark and capturing network traffic while browsing Instagram/YouTube for 2 minutes, the following protocols were commonly observed:
 - **TCP (Transmission Control Protocol)** – used for reliable data transfer
 - **DNS (Domain Name System)** – used to resolve domain names into IP addresses
 - **HTTPS (Hypertext Transfer Protocol Secure)** – used for secure web browsing and streaming
- **Top 3 protocols observed:**
 - TCP
 - DNS
 - HTTPS



2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

ANS :

- Ping www.spotify.com (Slower)

```
C:\Users\HARSH>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [151.101.107.42] with 32 bytes of data:
Reply from 151.101.107.42: bytes=32 time=17ms TTL=59
Reply from 151.101.107.42: bytes=32 time=17ms TTL=59
Reply from 151.101.107.42: bytes=32 time=17ms TTL=59
Reply from 151.101.107.42: bytes=32 time=17ms TTL=59

Ping statistics for 151.101.107.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 17ms, Average = 17ms
```

- Ping www.zomato.com (Faster)

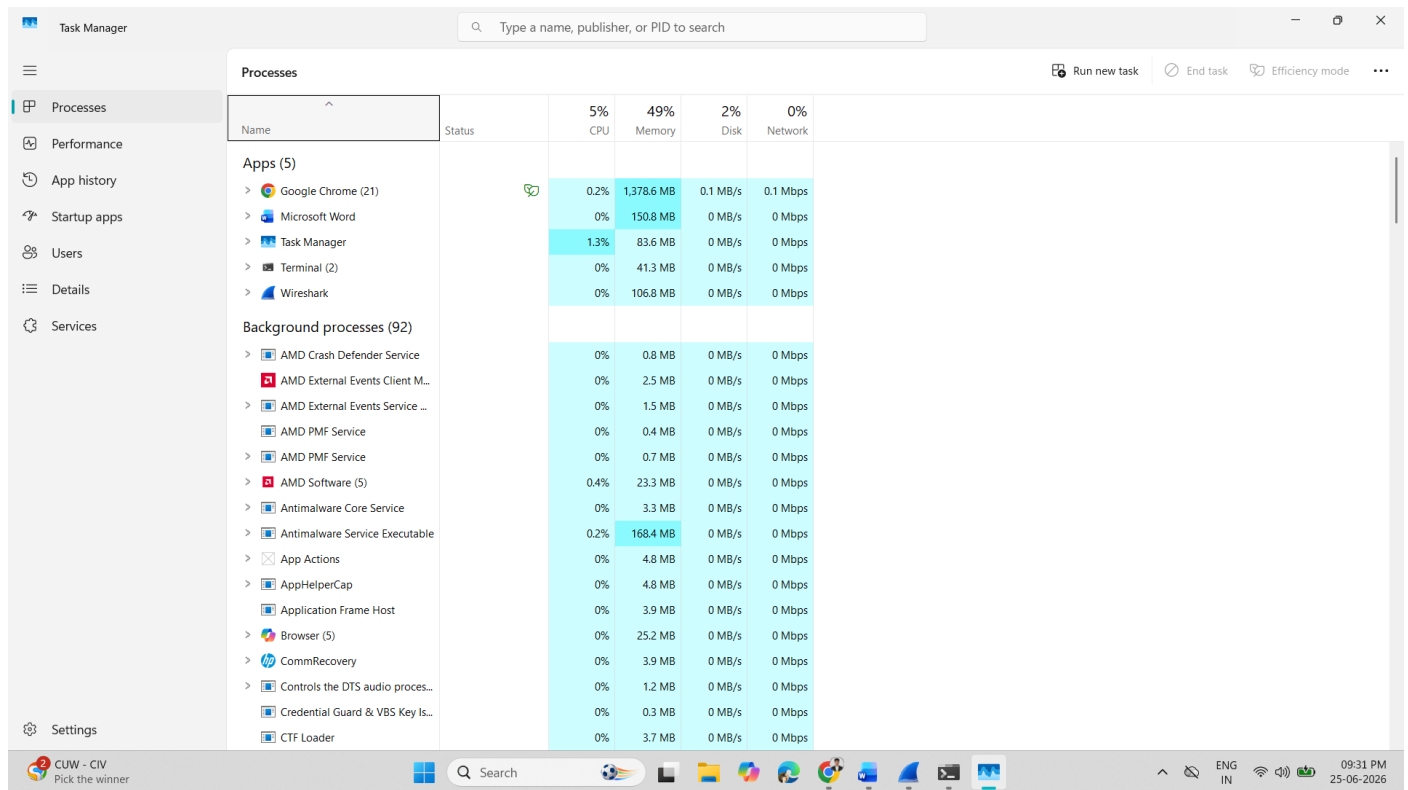
```
C:\Users\HARSH>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [49.44.140.145] with 32 bytes of data:
Reply from 49.44.140.145: bytes=32 time=12ms TTL=54
Reply from 49.44.140.145: bytes=32 time=13ms TTL=54
Reply from 49.44.140.145: bytes=32 time=12ms TTL=54
Reply from 49.44.140.145: bytes=32 time=13ms TTL=54

Ping statistics for 49.44.140.145:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 13ms, Average = 12ms
```

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

ANS :

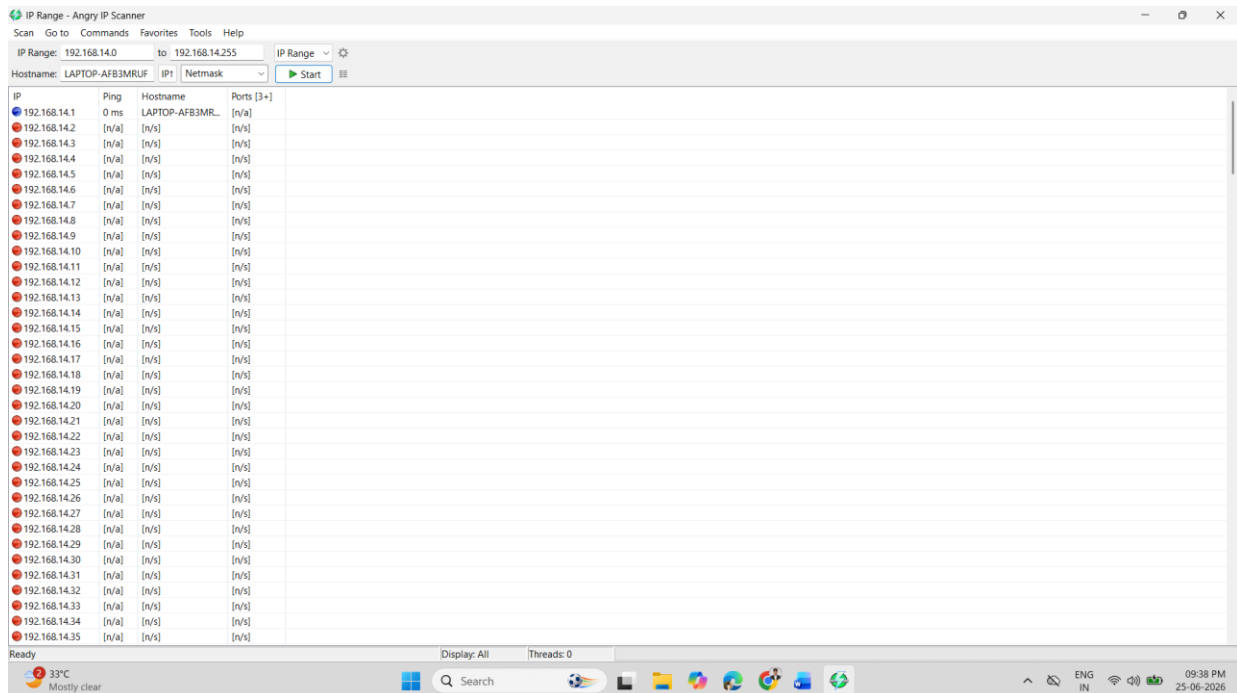


Name	Status	5% CPU	49% Memory	2% Disk	0% Network
Apps (5)					
> Google Chrome (21)		0.2%	1,378.6 MB	0.1 MB/s	0.1 Mbps
> Microsoft Word		0%	150.8 MB	0 MB/s	0 Mbps
> Task Manager		1.3%	83.6 MB	0 MB/s	0 Mbps
> Terminal (2)		0%	41.3 MB	0 MB/s	0 Mbps
> Wireshark		0%	106.8 MB	0 MB/s	0 Mbps
Background processes (92)					
> AMD Crash Defender Service		0%	0.8 MB	0 MB/s	0 Mbps
> AMD External Events Client M...		0%	2.5 MB	0 MB/s	0 Mbps
> AMD External Events Service ...		0%	1.5 MB	0 MB/s	0 Mbps
> AMD PMF Service		0%	0.4 MB	0 MB/s	0 Mbps
> AMD PMF Service		0%	0.7 MB	0 MB/s	0 Mbps
> AMD Software (5)		0.4%	23.3 MB	0 MB/s	0 Mbps
> Antimalware Core Service		0%	3.3 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0.2%	168.4 MB	0 MB/s	0 Mbps
> App Actions		0%	4.8 MB	0 MB/s	0 Mbps
> AppHelperCap		0%	4.8 MB	0 MB/s	0 Mbps
> Application Frame Host		0%	3.9 MB	0 MB/s	0 Mbps
> Browser (5)		0%	25.2 MB	0 MB/s	0 Mbps
> CommRecovery		0%	3.9 MB	0 MB/s	0 Mbps
> Controls the DTS audio proces...		0%	1.2 MB	0 MB/s	0 Mbps
> Credential Guard & VBS Key Is...		0%	0.3 MB	0 MB/s	0 Mbps
> CTF Loader		0%	3.7 MB	0 MB/s	0 Mbps

Process Name	Application	Memory Usage
chrome.exe	Google Chrome	800 MB – 1500 MB

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint : Make sure your firewall allows scanning, and only scan your own home/college network

ANS :



- 192.168.14.1 : LAPTOP-AFB3MRUF

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

ANS:

- **Cable Tester** – Used to check whether Ethernet cables are properly connected or damaged.
- **Crimping Tool** – Used to attach RJ45 connectors to network cables.
- **Multimeter** – Used to measure voltage, continuity, and detect electrical issues in network hardware.